Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years of experience building **education-focused full-stack products** across **Java**, **Python**, **JavaScript/TypeScript**, **SQL**, and cloud platforms. Strong in designing user-facing workflows, backend application logic, testing, debugging, and shipping scalable products that improve learning experiences.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University **Gandhinagar, In**
Bachelor of Technology in Computer Science and Engineering *November 2020 – May 2*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant AI education platform on **Azure SQL**, **PostgreSQL**, and cloud storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema, indexing, and query optimization.
- Built **FastAPI** backend services and validation-backed release workflows for **5K+ monthly jobs**, improving data reliability and production stability for instructor-facing experiences.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and shipped a **Flask + React** grading platform serving **150+ instructors**, with modular UI flows, secure file ingestion, rubric workflows, and **MySQL/AWS RDS**-backed APIs handling **10K+ submissions**.
- Implemented concurrent batch processing and **GitHub Actions CI/CD**, sustaining **10K+ grading requests/day** while cutting feedback turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software with microservice architecture and **20+ unit/integration suites** using **Jenkins**, PyTest, and Mockito, contributing tested code across **3 operating systems** in an Agile Scrum environment.
- Optimized API and ETL components using **OOP patterns**, **AWS Glue**, and S3, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** platform with **REST APIs**, **WebSockets**, and microservices, supporting **8+ business workflows** and **2M+ synchronized records** for a user-facing operations product.
- Automated deployments and validation with **Jenkins**, Docker, and Azure SQL-backed services, reducing deployment time by **50%** while reaching **95%+ successful build validation**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built full-stack booking workflows with **React**, **TypeScript**, Flask, and Firebase across **50K+ records**, covering reservations, pricing, payment status, and real-time synchronization.
- Integrated webhook-driven payment flows and **12+ backend integration points**, improving confirmation speed by **40%**, reducing manual payment handling by **70%**, and strengthening end-to-end reliability.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, Client-Server Architecture
Frontend: React.js, Next.js, HTML, CSS, TailwindCSS, D3.js
Databases: PostgreSQL, MySQL, Oracle-adjacent SQL workflows, SQL Server, MongoDB, Azure SQL
Cloud & Infrastructure: Linux, Apache, AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub Actions
AI/ML: LLM Systems, GPT-4o, Claude, OCR Workflows, Prompt-Oriented Development
Engineering Practices: CI/CD, Unit & Integration Testing, Debugging, Performance Optimization, Microservices, Agile, System Design

Projects

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, REST APIs*
- Built an AI document platform over **50K+ documents** with semantic search, summarization, and optimized retrieval pipelines, reducing latency by **40%** and supporting **200+ users**.

Chat2SQL | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, accelerating query authoring, schema exploration, and structured execution across a **Java + TypeScript** codebase.

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented **6 scheduling algorithms** with interactive controls and live visualizations across **1K+ simulated process combinations**, strengthening debugging, application logic, and user-facing tooling.